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In this analysis of my final project, I will discuss what the best model is and how I came to this conclusion, even though there were multiple models that outperformed one another in different parts of the testing. After all the individual models were trained, I used the F1 score metric, which balances precision and recall, and from previous modules I learned not always to trust the accuracy number of a model. Another thing I did to see the best model was to test both on validation and the test to see if models look good on both or just one.

When I first started this assignment, I believed that there would always be a clear cut model that could be way better or just slightly better than the others. What I did not expect, though, was multiple models to win in different circumstances. I saw this when I printed out my F1 scores; I first saw that random forest had the highest validation score out of all the models, but when I looked at the test split, it showed SVC as the best. Even though I made the same churn rate for each split, each customer is different, which could have made the models make different decisions. I then thought, why would these be the best models, and after thinking about it I figured out why. First, for the Random Forest, as discussed in Module 9, this has a wisdom of the crowd effect. This is very useful in this scenario, as each tree is individually voting whether the customer will churn or not, but because of the sheer number of trees, it has the wisdom of the crowd effect and can take full advantage and give an accurate model. Then I wondered why SVC did so well on this one as well; that is just because SVC is a stable model. Instead of going after each data point, it takes the average and draws a boundary. This works especially well with every feature being on the same scale.

In the data, most of the models landed around 0.60 to 0.64, but there was one thing I noticed: XGBoost and KNN both scored very poorly compared to the others. I then thought about why they would score poorly and came up with this reason. XGBoost was just undercut by its recall, but its precision was actually the highest; I was doing it off precision; it would win. KNN was bad on both ends; I reasoned this to be because of one hot encoding. I heard that KNN models don’t perform as well. I also compared the predictions with voting and Bayesian; they both landed very similarly, as both of them weighted each section at around 33%

In conclusion, the model I chose to be the best was the random forest. Not only does the wisdom of the crowd effect become much more than any individual model, but it also has the highest recall, which in a model like this, having a high recall is what you want, as even if the

customer only has a 10% chance to churn in the model's eyes, it's still worth flagging for the company.